

class 3



$$y = X\beta + u, \quad u \sim N(0, \sigma^2 I_n)$$

X 's fixed. $\text{rank}(X) = k$

$$b = (X'X)^{-1}X'y \quad \left. \begin{array}{l} y = \underbrace{Xb}_{\downarrow \substack{x(X'X)^{-1}X'y}} + \underbrace{(y - Xb)}_e \end{array} \right\} = P_X y + \underbrace{(I_n - P_X)}_{M_X} y = P_X y + M_X y$$



 projection matrix

$$P_X \cdot P_X = P_X \quad P_X' = P_X$$

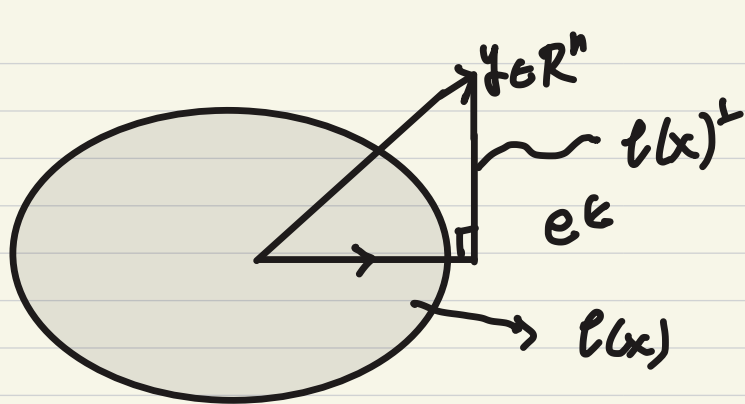
$$M_X \cdot M_X = M_X$$

idempotent
 symmetric

}

orthogonal projection

$$P_X X = X$$



$$\dim l(x) = k$$

$$+ \dim l(x)^\perp = n - k$$

$$P_X e = P_X (I - P_X) y = 0$$

$$M_X X = (I - P_X) X = X - X = 0$$

$$M_X e = M_X (I - P_X) y = (M_X - M_X P_X) y$$

$$= e$$

$$y = Py + My$$

$$y'y = (Py + My)'(Py + My)$$

$$= y'Py + y'My$$

T.T. S.S. = explained S.S. + Residual S.S.

$$R^2 = \frac{y'Py}{y'y}$$

$$\frac{y'Py}{y'y} = R^2 = 1 - \frac{y'My}{y'y}$$

$$y = x\beta + u$$

$$y_i = x_{i1}\beta_1 + \dots + x_{ik}\beta_k + u_i$$

$$\vdots$$

$$y_n = x_{n1}\beta_1 + \dots + x_{nk}\beta_k + u_n$$

$$R^2_{\text{adjusted}} = 1 - \frac{(y'My)/(n-k)}{y'y/n-1}$$

$$My = M(x\beta + u)$$

$$= Mx\beta + My = Mu$$

$$My = Mu \Rightarrow y'My = \underset{\substack{\uparrow \\ \text{RSS} = e'e}}{u'Mu}$$

$$u \sim N(0, \sigma^2 I_n)$$

$$E(e'e) = E(u'Mu) = E(\text{trace}(u'Mu)) = E(\text{trace}(u'uM)) = E(\text{trace}(Mu'u))$$

$$\text{trace}(A) = \sum_{i=1}^n a_{ii} \quad \text{trace}(AB) = \text{trace}(BA) = \text{trace}(CAB) = \text{trace}(C) \cdot E(u'u')$$

$$= \sigma^2 \text{trace}(I_n)$$

$$= \sigma^2 \text{trace}(I_n - P_x)$$

$$\text{trace}(M)$$

$$= \text{trace}(I_n - X(X'X)^{-1}X')$$

$$= \text{trace}(I_n) - \text{trace}(I_k)$$

$$= n - k$$

$$E(e'e) = \sigma^2(n - k)$$

$$s^2 = \frac{e'e}{n - k} \quad E(s^2) = E\left(\frac{e'e}{n - k}\right)$$

$$= \frac{\sigma^2(n - k)}{n - k}$$

Partitioned Regression Model

$$y = X\beta + u$$

$$X = \begin{bmatrix} X_1 & X_2 \end{bmatrix} \quad K = K_1 + K_2$$

$$n \times K \quad n \times K_1, n \times K_2$$

$$y = [x_1 \vdots x_2] \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_2 \end{pmatrix} \begin{matrix} k_1 \times 1 \\ k_2 \times 1 \end{matrix}$$

$$= x_1 \beta_1 + x_2 \beta_2 + u$$

$$b = (x'x)^{-1}x'y$$

$$= \left[(x_1 \vdots x_2)' (x_1 \vdots x_2) \right]^{-1} (x_1 \vdots x_2)' y = 0$$

$$= \left[\begin{pmatrix} x_1' \\ x_2' \end{pmatrix} (x_1 \ x_2) \right]^{-1} \begin{pmatrix} x_1' \\ x_2' \end{pmatrix} y$$

$$b = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} x_1' x_1 & x_1' x_2 \\ x_2' x_1 & x_2' x_2 \end{pmatrix}^{-1} \begin{pmatrix} x_1' y \\ x_2' y \end{pmatrix}$$

Frisch-Waugh-Lovell (FWL)

$$y = x_1 \beta_1 + x_2 \beta_2 + u \longrightarrow \ell(x_1) \longrightarrow \dim \ell(x_1) = k,$$

$$\longrightarrow P_1 = x_1 (x_1' x_1)^{-1} x_1'$$

$$\ell(x_1)^\perp$$

$$\dim \ell(x_1)^\perp = n - k,$$

$$x_1 \subset x$$

$$p_1 x_1 = x_1 \quad p x_1 = x_1 \quad m_1 x_1 = 0$$

$$m_1 e_1 = e_1$$

$$m_1 m = m$$



because m is smaller

$p p_1 = p_2 \rightarrow$ because p_1 is smaller.